

Chengyi Cai

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EDUCATION

Carnegie Mellon University (CMU) , H. John Heinz III College	May 2026
Master of Science, <i>Public Policy & Management–Data Analytics</i>	GPA: 4.04/4
Jinan University (JNU) <i>Exchange at Xiamen University (Fall 2023)</i>	July 2024
Bachelor of Law, <i>Int. Affairs and Int. Relations</i> ; Bachelor of Arts, <i>Japanese</i>	GPA: 3.94/4
Cambridge University Summer Research Programme: <i>International Relations and Governance</i>	August 2023

SKILLS & COURSEWORK

Technical: Python (scikit-learn, PyTorch, Transformers), R, SQL, SPSS, ArcGIS, Tableau, Datawrapper, Arduino, MongoDB, Microsoft
Courses: Machine Learning, Big Data Science, Engineering Optimization, AI Engineering and Policy for Climate Mitigation, Applied Econometrics, Database Management, Statistics & Modeling, Geographic Information System, Finance, Sociology & Anthropology

RESEARCH EXPERIENCE

CMU - Scott Institute for Energy Innovation , Research Assistant (2-person team)	May 2025 - Present
China Industrial Heating Decarbonization Technologies	Advisor: Valerie Karplus

- Built Excel-based LCOH (CAPEX/OPEX) models and a process-level lifecycle assessment (LCA) for renewable natural gas (RNG) and industrial heat pumps; quantified CO₂-equivalent emissions and cost competitiveness relative to conventional fuels.
- Quantified RNG feedstock potentials in China; compared global RNG certification and tracking schemes and proposed staged policy recommendations for Apple's supply-chain decarbonization and for refining China's CCER biomethane methodology.
- Authored a 22-page technical report and co-authored a policy memo on China's RNG potential and the global policy landscape; findings were presented at the [7th Corporate Carbon Neutrality Roundtable](#) co-hosted by Tsinghua University and CMU.

CMU - Heinz College , Research Assistant (Independent study)	May 2025 - Present
Renewables Optimization & AI Workload Scheduling	Advisor: Peter Zhang

- Developed a bi-level optimization framework to align AI datacenter (DC) workloads with renewable energy curtailment; integrated robust spatiotemporal scheduling considering climate disruption uncertainty; presented results on the [CSUN Energy Systems seminar](#).
- Crafted a two-stage robust optimization for datacenters' long-term site selection and daily load scheduling under unpredictable outages.
 - Built a day-ahead linear programming (LP) demo that routes flexible DC workloads across sites × hours to minimize energy price, grid carbon, latency, curtailment use, and peak penalties; ran synthetic cases with ISO/EIA/EPA inputs and tested sensitivities.
 - Assembled the uncertainty set, curated 4k+ DOE-417 outage records of 2002-2023 and compiled a DC energy-profile table (onsite generation/storage mixes with expenditures, build times, climate resilience) to parameterize the model for disruptions.

Energy and Climate Studies	May 2021 - Present
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- Co-developed a two-stage optimization model for EV fast-charger upgrades in Shenzhen (MIP siting + convex operations with battery degradation; implemented in Gurob and CVXPY), and ran sensitivity analyses to stress-test planning decisions.
- Conducted a GIS-based multi-criteria analysis for U.S. data center siting, combining 5 county-level indicators (renewables share, disaster risk, grid reliability, prices, etc.) into a composite index and multivariate clusters, delivered as an interactive [StoryMap](#).
- Designed a GenX-inspired predict-then-optimize (PTO) pipeline for [CAISO Generator 2030 Planning](#), that forecasts renewables and clusters representative days, co-optimized capacity & dispatch; ran sensitivity analyses across carbon-price scenarios (Python/Gurobi).
- Authored a policy brief and memo on West Virginia's energy just transition and economic diversification; published findings in a [Shorthand web report](#) integrating narrative text and interactive visualizations using Tableau and Datawrapper.
- Collected and analyzed panel data from 30 Chinese provinces (1996–2020) to assess the causal relationship between air pollution and mortality rates by developing multiple econometric linear regression models by Python.
- Explored U.S. lung cancer trends with a focus on Kentucky through data analysis and performed regression modeling using Python, to identify key predictors like smoking rates and demographic factors, and provided public health recommendations.
- Collaborated in an international team of 5 at MIT Policy Hackathon 2023 to establish an [API Ecosystem](#) for EO data access and water management in East Asia; labeled 500+ policy texts (AU Agenda 2063) to create expert-annotated datasets for UNDP AI solutions.
- Completed multiple international-relations and climate-diplomacy research papers (EU energy transitions under the Russia-Ukraine Conflict, Japan-led Asian ESM network, climate-induced migration in Kiribati, UK environmental diplomacy), executed multi-language literature review, data collection via government websites.
- **Bachelor's thesis** (Advisor: Yuheng Fu): Proposed a "Cooperative Will and Ability" framework for UNFCCC climate games, using 1990–2020 socioeconomic and energy data to quantify countries' incentives to cooperate (perceived emission-reduction benefits) and abilities to act (GHG share and development constraints) for major emitters (US, China, EU, Russia) .
- **Independent Research** (Advisor: Anqi Hu): Secured national funding for a year-long sociology project on regional inequality in Jiangsu; conducted fieldwork in 3 cities (17 in-depth interviews), and authored a 39-page report plus a 12-page paper with a new 'Regional Identity and Space Status' framework, producing engaging [video](#) and [tweets](#) to enhance the public accessibility.

TECHNICAL PROJECTS

Deep Learning & Natural Language Processing	January 2025 - Present
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- Designed and implemented a frame-level [speech recognition model](#) using PyTorch, built a Deep Pyramid Network with 7 fully connected layers (BatchNorm, Dropout) and optimized using AdamW and CosineAnnealingLR, achieving 0.85617 validation

accuracy within 20M parameters.

- Applied transformer embeddings and BERTopic to a Chinese climate-adaptation petitions dataset, conducted a customized stopwords preprocessing, tuned model hyperparameters and produced topic-hierarchy visualizations in a 6-page report showing regional differences with policy recommendations for geographically targeted flooding-adaptation investments.

Machine Learning & Programming

September 2024 - Present

- Developed a computer-vision pipeline in a 5-member team to extract body/keypoint measurements from images, applied K-means clustering to identify female body-type groups, and produced clustered anthropometric templates for [women's PPE sizing](#) with 3D printing for physical validation, yielding a usable prototype and design artifacts for manufacturer dialogue.
- Trained and compared 10 ML models (RF, XGBoost) to predict Allegheny County's [311 system service-request closure times](#) with feature engineering, rolling-window cross-validation and iteratively pruned features, aiming at equity-aware policy interventions.
- Built predictive poverty models from geospatial features using ArcGIS and Python, combining nightlight intensity, population data, and regularized regressions to generate interpretable regional poverty estimates.
- Developed a data-driven [apartment recommendation tool](#) utilizing web scraping and Google API to gather data on rental options, transit accessibility, neighborhood safety, and users' personalized inputs, with geographic visualizations using Folium.

Human-Computer Interaction

August 2024 - August 2025

- **AR for Public Speaking:** Tested 3 VR stress-reduction techniques for public presentation; conducted participant interviews and usability assessments, provided design recommendations to improve UX and stress-mitigation workflows for the application.
- **EEG-based Brain-Computer Interface(BCI) Research:** Collaborate on a [PhD project](#) using EEG signals for non-invasive BCI applications; provide initial data through imagined robotic hand movement exercises, contributing to modeling with 70% accuracy rate.
- **Robotic Hand Teleoperation Research:** Contributed to a PhD project on two teleoperation methods (Kinematic Twin & Visual Tracking Leap Motion) for remote robotic arm control; assessed accuracy in object manipulation for future usability improvements.

PROFESSIONAL EXPERIENCE

Consulate General of Sri Lanka in Guangzhou, Administrative Assistant Intern

February - May 2024

- Processed ~10 visa applications per week, crafted diplomatic correspondences and translated/vetted tender documents and visit notes, including individual simultaneous interpretation for the Commercial Counselor during a 2-day Future Business Forum visit.

China Blue - Marine Conservation NGO, Research Assistant Intern

July 2023 - June 2024

- Served as project secretary for the [Climate Change sub-forum](#) at the 2023 China Ecological Civilization Annual Conference and COP28 China Pavilion Side Event 9; coordinated 10+ NGO leaders and scholars and engaged 200+ in-person attendees.
- Co-organized a Regional NGO Cooperation [Forum for Sustainable Coastal Communities](#) (partners incl. JARI, RARE, NCC, wetland institutes in Southeast Asia); hosted 100+ participants and distilled community-led coastal-governance practices.
- Examined 5 Waste-Free City cases under UNDP, leading to a 13-page report to guide Haikou pilot community growth; produced literature syntheses on food waste, methane, and composting; contributed quarterly briefs on marine conservation in Southeast Asia.

MP Management Professional Group, Government Consulting Intern

September 2023 - February 2024

- Performed data cleaning and analysis on a subsidy funds dataset (2016-2020) from the Culture, Tourism, and Sports Bureau of Guangzhou, presenting findings through informative charts produced by Excel and SPSS.
- Interviewed district bureaus and 18 industrial representatives in field research, enhancing understanding of regional industrial dynamics for the development of Dagang Town, Guangzhou; analyzed industry data, including scale, ecological benefits, and tax contributions, to assess the feasibility of future industry development, based on statistical data and local government reports.

LEADERSHIP EXPERIENCE

CMU Student Using Data for Social Good, Project Manager

October 2024 - January 2025

- Managed a 4-person team and delivered a 15-page data analysis report for nonprofit "A Caring Place," conducting EDA, demographic analysis of child abuse cases, and policy-oriented visualizations to improve program targeting and service delivery.

CMU Robotics Club 2024 RED ROBOT Hackathon, Mentor

November 2024

- Self-taught base robot assembly and Arduino programming to mentor hackathon teams; assisted with code deployment, debugging, and hardware assembly during competition to improve team reliability and accelerate iteration cycles.

Green Youth Volunteer Service Team, Deputy Leader

September 2020 - July 2022

- Led a team of 73 students, oversaw departmental operations and funds management, fostered external partnerships with organizations (e.g., Tencent Public Welfare), enhancing weekly meeting efficiency and upgrading Outreach branch into Data-Outreach integration.
- Conceptualized with 2 leaders and executed over 11 initiatives such as clothing recycling (500+ participants) while operating an online community individually and facilitating group learning (engaging 180+ participants posting 1300+ tweets in one week).
- Facilitated weekly education modules at 3 local elementary schools for 2 years, focusing on subjects like the Zero Carbon Behavior.

HONORS & AWARDS

1st Prize (\$500), 2025 Engineering Design, Innovation & Entrepreneurship (EDIE) Makeathon - Personalized women's PPE

3rd Prize, 2024 CMU Student IronViz - [Tableau Dashboard on candy & mental health](#)

Top-15 Finalist, 2024 Hack-a-Startup, CMU Tepper Business School - [Hosting LLM tools for agent customization](#)

MSPPM-DA Scholarship, CMU Heinz College, 2024-2026

China National Scholarship (Nationwide Top 0.2%), 2022; JNU Third-Class Scholarship (College-wide Top 3%), 2021

National Project for Undergraduate Innovation and Entrepreneurship, 2022 - Research Funds & Excellent Completion (Individual)